Dmitrii Zagorulia

Machine Learning for Scientific Discovery

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Education

Moscow Institute of Physics and Technology (MIPT) *Dolgoprudny, Russia*

MSc in Computer Science and Computer Engineering *2024 – 2026*

- Specialty in *Scientific Programming*.

MIPT *Dolgoprudny, Russia*

BSc in Applied Mathematics and Physics *2020 – 2024*

- Specialty in *Theoretical Physics & Astrophysics* joint with P. N. Lebedev Physical Institute (LPI) RAS.

Research Experience

P. N. Lebedev Physical Institute RAS *Moscow, Russia*

Advisor: Prof. Mikhail Lisakov (Pontifical Catholic University of Valparaíso) *2023 – PRESENT*

- Developed supervised, self-supervised, and transfer learning models in *PyTorch* for morphological classification of parsec-scale AGN jets in Astrogeo VLBI FITS images.
- Generated over 200 000 synthetic Blandford–Königl jet images, preprocessed more than 130 000 observations from the Astrogeo database, and trained a ResNet-18 with SIDDA domain adaptation, achieving 86% accuracy on real VLBI images.
- *Master's thesis*: "Domain Adaptation of Simulation Data for Morphological Classification of Relativistic Jets"
- *Bachelor's thesis*: "Morphological Classification of Active Galactic Nuclei"

Institute of Astronomy and Astrophysics, Academia Sinica *Taipei, Taiwan (hybrid)*

Advisors: Dr. Hsien Shang, Dr. Ruben Krasnopolsky *2021 – 2025*

- Derived analytical solutions for the compressed ambient medium and stellar wind, matched across the contact discontinuity.
- Implemented an exact planar Riemann solver for the one-dimensional Euler equations to model shock structures.
- Studied self-similar stellar-wind bubbles and developed a numerical solver for the hydrodynamic system.
- Published the resulting work as a first-author article in *PASP*.

Special Astrophysical Observatory RAS *Nizhnij Arkhyz, Russia (hybrid)*

Advisor: Prof. Grigorii Beskin *2021 – 2024*

- Investigated the transition between quasi-spherical and disc accretion for supersonic isolated stellar-mass black holes moving through molecular clouds.
- Analysed black hole population-synthesis data and obtained a joint probability map of black-hole mass and velocity.

MIPT *Dolgoprudny, Russia*

Advisor: Prof. Vasily Beskin *2021 – 2023*

- Modelled the electric potential above the polar caps of orthogonal interpulse pulsars in the vacuum approximation, finding non-stationary secondary-particle production.
- Constructed synthetic mean radio profiles for eight pulsars while accounting for the non-axisymmetric plasma outflow.
- Developed algorithms for analysing the kinetic equation governing the neutron-star distribution function.

Publications

D. S. Zagorulia, H. Shang, R. Krasnopolsky. Polytropic Wind-Driven Bubbles and their Shock Structures in Radially Stratified Ambient Media. *PASP*, 137, 064301 (2025). DOI: 10.1088/1538−3873/add80e

D. S. Zagorulia, M. M. Lisakov. Classification of Jets in Active Galactic Nuclei using Machine Learning. *Modern Astronomy: From the Early Universe to Exoplanets and Black Holes*, 131-135 (2024). DOI: 10.26119/VAK2024.018

D. S. Zagorulia, G. M. Beskin. Sphere or Disk? Accretion of Interstellar Gas onto Isolated Stellar-Mass Black Holes. *Modern Astronomy: From the Early Universe to Exoplanets and Black Holes*, 599-603 (2024). DOI: 10.26119/VAK2024.095

V. S. Beskin, **D. S. Zagorulia**, A. Y. Istomin. Magnetic Fields of Neutron Stars. *Astronomy Letters*, 47, 686–694 (2021). DOI: 10.1134/S1063773721100017

Engineering and Applied ML Experience

Yandex *Moscow, Russia*

Machine Learning Engineer, Voice Input Service (full-time, on-site) *Apr. 2026 – Jun. 2026*

- Trained CatBoost models on audio-quality features to automate side-by-side evaluation and reduce human-assessment cost.
- Implemented and evaluated quantization-aware training for audio models.
- Analyzed firmware parameters of a new device and their impact on input audio quality.

Scientific Programming Centre *Dolgoprudny, Russia*

Computer Vision Engineer (part-time, remote) *Feb. 2026 – Jun. 2026*

- Developed a real-time computer vision system for flotation monitoring: YOLOv11-based bubble segmentation from RTSP streams and extraction of bubble-size and distribution metrics.
- Implemented a module for recognizing digital readings from industrial camera footage.
- Built an IP camera control module for Hikvision devices using ISAPI and RTSP.

Laboratory of Intelligent Robotics Technologies, MIPT *Dolgoprudny, Russia*

Robotics Software Engineer (full-time, on-site) *Jun. 2025 – Mar. 2026*

- *Robotic Arm*: Developed ROS2 software and a C++/pybind11 backend for communication with robotic-hand sensors and motors over RS485/Modbus.
- *Warehouse Robot*: Containerized and debugged an ORB-SLAM3 pipeline for real-time room mapping and robot localization against a prebuilt map, using gyroscope data for improved positioning.

Teaching Experience

ML4Astro, MIPT *Dolgoprudny, Russia*

Lecturer *2026*

- Designed and taught an undergraduate elective on modern machine-learning methods for astrophysics, including physics-informed learning, probabilistic inference, domain adaptation, uncertainty quantification, and interpretability.

Olympiad Astronomy, School №2107 *Moscow, Russia*

Teacher *2024 – 2026*

- Conducted weekly workshops, covering the fundamentals of spherical astronomy, celestial mechanics, astronomical optics, astrophysics, and cosmology. Created problem sheets tailored to each workshop.

Algorithms and Data Structures, MIPT *Dolgoprudny, Russia*

Teaching Assistant *2023 – 2025*

- Led weekly problem-solving sessions, developed programming contests, and advised students on algorithm implementation.

Observing Time Awarded

Co-Investigator (PI: R. Rao), *Submillimeter Array* Proposal 2024B-S045 – 4 hours awarded.

Skills

Programming	Python (PyTorch, Lightning, CatBoost, scikit-learn, NumPy, SciPy, Astropy), C/C++ (CMake, Boost, MPI)
Astronomy Software	Difmap, CASA, CARTA
Languages	Russian (native), Chinese (fluent), English (fluent), French (elementary)

Presentations

** presenting author; + mentored*

D. S. Zagorulia, M. M. Lisakov. 2026. Simulation-to-Real Learning for VLBI Jet Morphology Classification. Oral presentation: IAIFI Summer Workshop, Cambridge, MA.

D. S. Zagorulia, M. M. Lisakov. 2026. Simulation-to-Real Learning for VLBI Jet Morphology Classification. Poster presentation: AI4Physics Workshop at the International Conference on Machine Learning, Seoul, South Korea.

D. S. Zagorulia, M. M. Lisakov. 2026. Morphological Classification of Jets in Active Galactic Nuclei. Poster presentation: European Astronomical Society Annual Meeting, Lausanne, Switzerland.

D. S. Zagorulia, M. M. Lisakov. 2026. Morphological Classification of VLBI Images of AGN Jets Using Contrastive Learning. Oral presentation: LPI-MIPT Observational Astrophysics Seminar, Moscow, Russia.

D. S. Zagorulia, M. M. Lisakov. 2026. Morphological Classification of Jets in Active Galactic Nuclei. Oral presentation: 23rd Conference of Young Scientists, Space Research Institute RAS, Moscow, Russia.

I. P. Yakushenoks*+, **D. S. Zagorulia**. 2026. Redshift Estimation of Galaxies Using Convolutional Neural Networks. Oral presentation: 23rd Conference of Young Scientists, Space Research Institute RAS, Moscow, Russia.

D. S. Zagorulia, M. M. Lisakov. 2026. Morphological Classification of Jets in Active Galactic Nuclei. Oral presentation: Astroinformatics, Cologne, Germany.

D. S. Zagorulia, H. Shang, R. Krasnopolsky. 2025. Polytropic Wind-Driven Bubbles and their Shock Structures in Radially Stratified Ambient Media. Oral presentation: All-Russian School-Conference of Young Astronomers, Pushkov Institute of Terrestrial Magnetism, Ionosphere and Radiowave Propagation RAS, Troitsk, Russia.

D. S. Zagorulia, H. Shang, R. Krasnopolsky. 2025. Polytropic Wind-Driven Bubbles and their Shock Structures in Radially Stratified Ambient Media. Oral presentation: 22nd Conference of Young Scientists, Space Research Institute RAS, Moscow, Russia.

D. S. Zagorulia, M. M. Lisakov. 2025. Morphological Classification of Jets in Active Galactic Nuclei. Oral presentation: AstroAI lunch talk, Center for Astrophysics | Harvard & Smithsonian, Boston, MA.

D. S. Zagorulia, M. M. Lisakov. 2024. Morphological Classification of Jets in Active Galactic Nuclei. Oral presentation: Active Galaxies at Different Scales and Wavelengths, Special Astrophysical Observatory RAS, Nizhnij Arkhyz, Russia.

D. S. Zagorulia, M. M. Lisakov. 2024. Morphological Classification of Jets in Active Galactic Nuclei. Poster: All-Russian Astronomical Conference, Special Astrophysical Observatory RAS, Nizhnij Arkhyz, Russia.

D. S. Zagorulia, G. M. Beskin. 2024. Sphere or disk? Accretion of Interstellar Medium on Isolated Stellar-mass Black Holes. Poster: All-Russian Astronomical Conference, Special Astrophysical Observatory RAS, Nizhnij Arkhyz, Russia.

D. S. Zagorulia, M. M. Lisakov. 2024. Morphological Classification of Jets in Active Galactic Nuclei. Oral presentation: 8th International Conference on Deep Learning in Computational Physics, M. V. Lomonosov Moscow State University, Moscow, Russia.

D. S. Zagorulia, M. M. Lisakov. 2024. Morphological Classification of Jets in Active Galactic Nuclei. Oral presentation: 66th International MIPT Scientific Conference, MIPT, Dolgoprudny, Russia.

D. S. Zagorulia, M. M. Lisakov. 2023. Morphological Classifier of Active Galactic Nuclei. Poster: High Energy Astrophysics Today and Tomorrow, Space Research Institute RAS, Moscow, Russia.

V. D. Chernosov*, V. I. Tsurkis, **D. S. Zagorulia**, V. S. Beskin. 2023. To Morphology of Mean Profiles of Orthogonal Interpulse Pulsars. Poster: Physics of Neutron Stars, Ioffe Physical-Technical Institute RAS, Saint-Petersburg, Russia.

D. S. Zagorulia, V. D. Chernosov, V. I. Tsurkis, V. S. Beskin. 2022. Orthogonal Interpulsar Pulsars as a Test of the Law of Evolution. Poster: High Energy Astrophysics Today and Tomorrow, Space Research Institute RAS, Moscow, Russia.

D. S. Zagorulia, A. Y. Istomin, V. S. Beskin. 2021. Magnetic Fields of Radio Pulsars. Oral presentation: 64th International MIPT Scientific Conference, Space Research Institute RAS, Moscow, Russia.

Outreach & Professional Development

Outreach

2024 **Cosmic Wednesday @ MIPT.Radio**, Founder & Podcast Co-host (6 episodes)

Professional Development

IAIFI Summer Workshop, Massachusetts Institute of Technology, MA, Jul. 2026.

IAIFI Summer School, University of Massachusetts, MA, Jul. 2026.

The 20th Heidelberg Summer School, University of Heidelberg, Germany, Sep. 2025. Attended a program on *AI in Astronomy*, including *Domain Adaptation*, *Simulation-based Inference*, and *Generative Modelling*.

Zwicky Transient Facility Summer School, University of Minnesota, remote, Jul. 2025. Completed hands-on training in *time-domain astronomy*, covering *ZTF Alert Stream Processing*, and *Rubin/LSST Filtering*.

AstroAI Workshop, Center for Astrophysics | Harvard & Smithsonian, remote, Jul. 2025. Participated in lectures and tutorials on *AI/ML in Astrophysics*, including *Image Reconstruction* and *Probabilistic Inference*.

Submillimeter Array Interferometry School, Center for Astrophysics | Harvard & Smithsonian, Hilo, HI, Jan. 2025. Led a 3-member team in preparing an observing proposal for the pre-stellar core L1551-MC. Trained in *Interferometry*, *Calibration*, and *Imaging* with CASA; analysed SMA & JCMT data of L1551-MC.

International Internship Pilot Program NSTC, Institute of Astronomy and Astrophysics, Academia Sinica, Taipei, Taiwan, Jul.–Aug. 2024. Completed lectures on different fields of *Astronomy*, and conducted research on hydrodynamic bubbles [*Zagorulia et al. 2025*].

Summer Program in Radio Astronomy, Pushchino Radio Astronomy Observatory RAS, Pushchino, Russia, Jul. 2022. Co-organized a 10-student program, attended lectures on *FRBs* and *VLBI*, and mapped jets in 3C 273 and M87 using Difmap.